Signal Filtering: Results

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1 Goal

Compare four filters (moving average, low-pass, complementary and Kalman) on the same noisy sine wave. Because the data is synthetic, the true signal is known, so each filter can be scored exactly. The filters will later be used on real MPU-6050 data in a separate project.

2 Setup

Data.

- True signal: $2\sin x$, with 101 points from 0 to 2π (step $\Delta x = 0.0628$ rad).
- Noisy position (like an accelerometer): true signal + uniform noise of width 1.0.
- Rate (like a gyroscope): $2\cos x$ + a constant bias + a little noise.
- Calibration: 200 readings taken while "still", used to find the bias, R and Q.

Filter settings.

- Moving average with $N = 10$ and low-pass with $\alpha = 0.82$. These two were chosen to have the same lag in theory, so they can be compared fairly.
- Complementary with $\beta = 0.975$.
- Kalman, with the bias, R and Q found from calibration.

Metrics.

- **RMSE:** the average size of the error between the filtered and true signal. Lower is better.
- **Lag:** how far the filtered signal trails behind the true signal.
- **Residual noise:** how much noise is left once the lag is removed.
- **Attenuation:** how tall the filtered wave is compared with the true one (1 = full height).

3 Results

All filters were run on the same data (random seed 42).

Table 1: Results for seed 42.

Signal	RMSE	Lag (rad)	Residual noise	Attenuation
Raw (no filter)	0.297	0.000	0.297	1.168
Moving average ($N = 10$)	0.402	0.314	0.105	0.999
Low-pass ($\alpha = 0.82$)	0.378	0.251	0.121	0.988
Complementary ($\beta = 0.975$)	**0.059**	0.000	**0.059**	0.983
Kalman	0.095	0.000	0.082	0.965

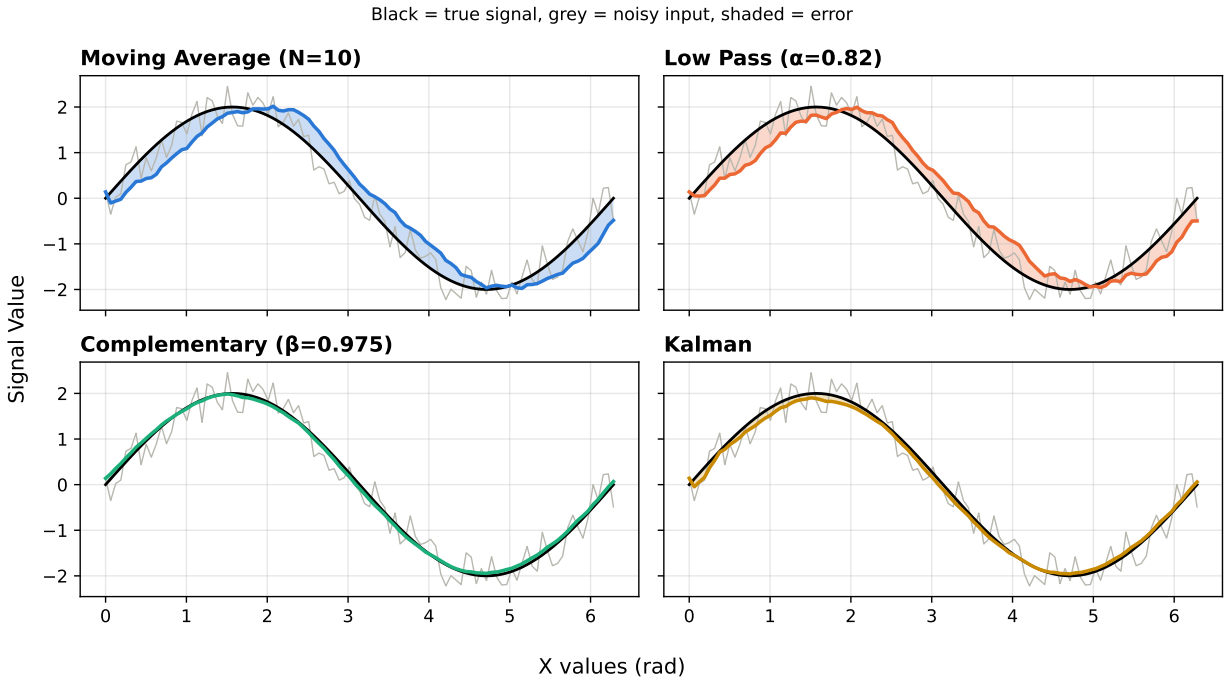


Figure 1: Each filter in its own panel. Black is the true signal, grey is the noisy input and the coloured line is the filter's output. The shaded area between the coloured line and the black line is the filter's error.

What Figure 1 shows. All four panels use the same axes, so they can be compared directly.

- **Moving average and low-pass (top row):** the coloured curve is smooth but sits to the *right* of the black line, so it is *late*. When the signal is rising the filter reads too low, and when it is falling it reads too high. That gap is the large shaded area, up to about 0.6 at its widest.
- **Complementary and Kalman (bottom row):** the coloured curve sits almost on top of the black line, so there is very little shading. These filters are smooth *and* on time.

Main finding. The moving average and low-pass filters remove most of the noise (residual noise drops from 0.30 to about 0.11), but they are so late that their RMSE is *worse* than doing nothing (0.38–0.40 vs 0.30). The complementary and Kalman filters use the rate to predict where the signal is going, so they remove the noise *without* being late.

4 Theory vs. Measured

The formulas below are the ones from my notes. σ is the noise level of the raw signal; for uniform noise of width w, $\sigma = w/\sqrt{12} = 0.289$.

Table 2: Theoretical values against measured values.

What	Formula	Theory	Measured (seed 42)
Raw noise	$w/\sqrt{12}$	0.289	0.297
Moving average lag	$(N-1)/2$ samples	4.5 samples	5 samples
Low-pass lag	$\alpha/(1-\alpha)$ samples	4.6 samples	4 samples
Moving average noise	$\sigma/\sqrt{N}$	0.091	0.105
Low-pass noise	$\sigma\sqrt{(1-\alpha)/(1+\alpha)}$	0.091	0.121
Calibrated R	$w^2/12$	0.0833	0.0802
Calibrated bias	true bias	0.1802	0.1816
Final Kalman gain K vs. $1-\beta$	—	0.025	0.026

Why the small differences?

- **Lag** is measured in whole samples, so a theoretical 4.5 or 4.6 can only show up as 4 or 5. Both are within one sample of theory.
- **Low-pass noise** comes out a little high for the same reason: its lag of 4.6 samples can only be removed as 4 whole samples, so the leftover shift is counted as noise.
- **Raw noise, R and the bias** differ slightly because they are estimated from a limited number of random samples.
- **Kalman gain:** the Kalman filter worked out its own gain from R and Q and ended up at 0.026, almost exactly the $1-\beta = 0.025$ I chose by hand for the complementary filter.

5 How the Kalman Gain Changes

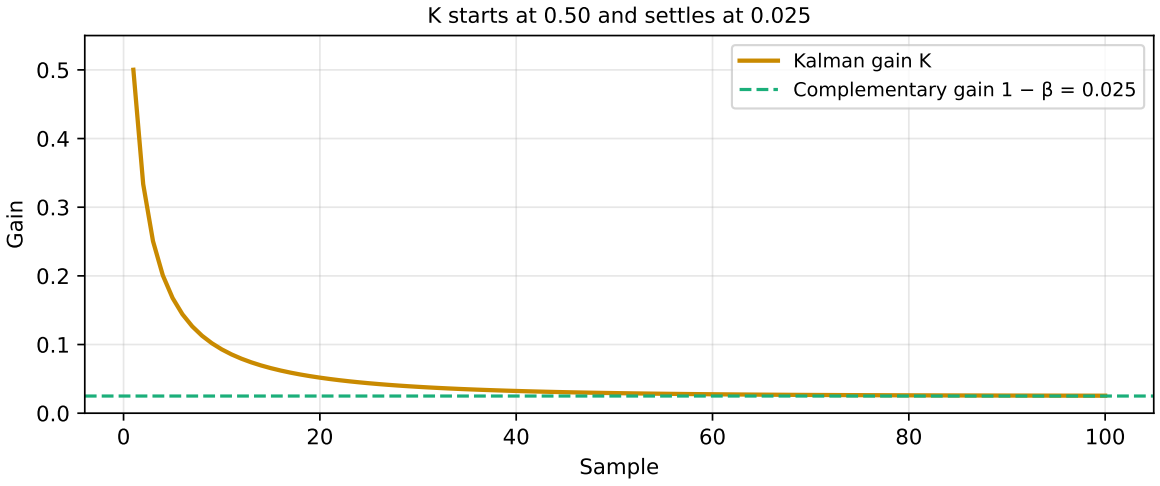


Figure 2: Kalman gain K at each step (yellow) and the complementary filter's fixed gain $1-\beta$ (green dashed).

What Figure 2 shows. The gain is the weight given to each new measurement. The Kalman filter starts with $K = 0.5$: at the beginning it is unsure of its estimate, so it trusts the measurements a lot. As it gains confidence, K drops and settles at about 0.026. The complementary filter uses the same small gain (0.025) from the very first sample.

6 Repeating the Test with 50 Different Noise Patterns

One run uses one random noise pattern, which could be lucky or unlucky. So the whole test was repeated with 50 different random seeds (0 to 49) and the results were averaged.

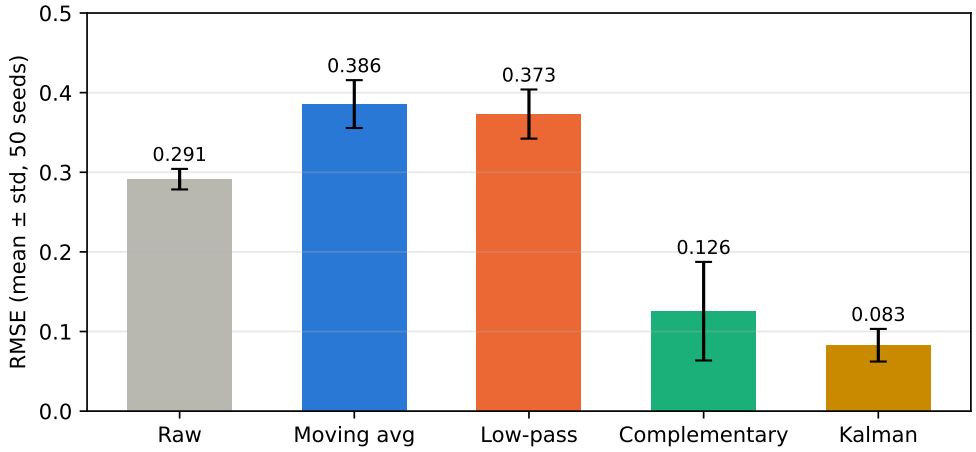


Figure 3: Average RMSE over 50 runs. The bar is the average; the thin line shows how much it varies between runs.

What Figure 3 shows. On average, the Kalman filter has the lowest error (0.083) and varies the least between runs. The complementary filter is second (0.126) but varies a lot. The moving average and low-pass filters are worse than the raw signal in all 50 runs.

Why this differs from seed 42. With seed 42 the complementary filter came first, but over 50 runs the Kalman filter wins 36 times. The reason is how they start. Both filters begin from the first noisy reading. If that reading is badly off:

- the complementary filter fixes it slowly, because its gain is always small (0.025);
- the Kalman filter fixes it within a few samples, because its gain starts large (0.5), as Figure 2 shows.

In seed 42 the first reading happened to be quite accurate, so the complementary filter's slow start did not hurt it.

7 Conclusions

1. **Moving average and low-pass filters remove noise but add lag.** On a changing signal, the lag costs more than the noise they remove, so their total error is worse than no filter.
2. **Using the rate removes the lag.** The complementary and Kalman filters predict where the signal is going, so they smooth the signal without falling behind. They cut the error by about 55–70%.

3. **The Kalman filter is the most reliable.** It found almost the same gain I picked by hand, and its large starting gain lets it recover quickly from a bad first reading.

4. **The results agree with theory**, within one sample for lag and within a few percent for noise.

Future work: test other signal speeds and filter settings, test a sudden step, and apply the filters to real MPU-6050 data.